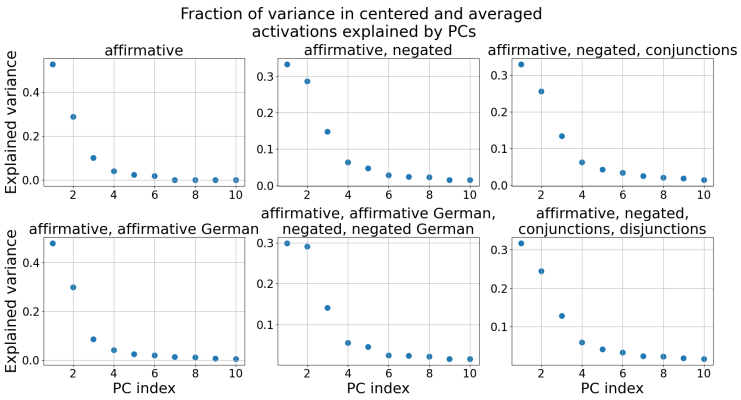


Figure 30: Gemma-2-27B: The fraction of variance in the centered and averaged activations $\tilde{\boldsymbol{\mu}}_i^+$, $\tilde{\boldsymbol{\mu}}_i^-$ explained by the Principal Components (PCs). Only the first 10 PCs are shown.

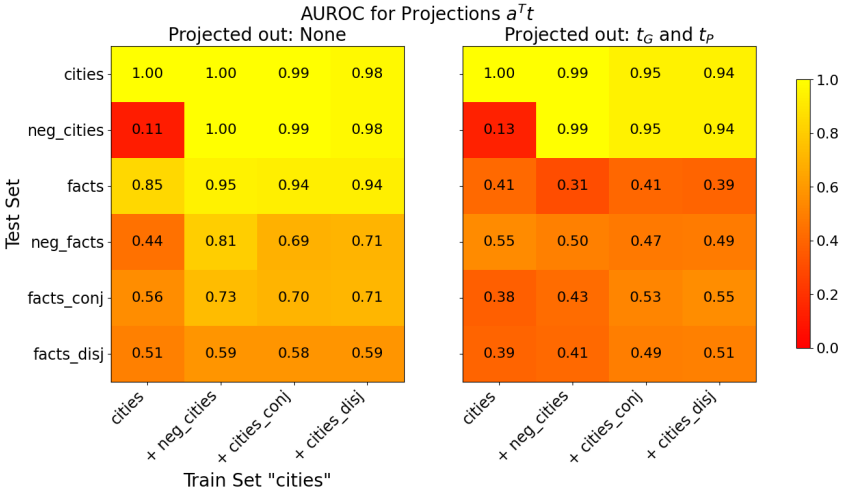
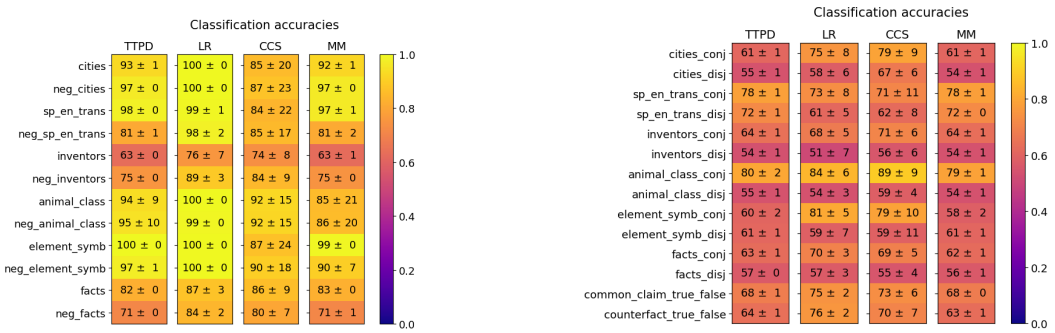


Figure 31: Gemma-2-27B: Generalisation accuracies of truth directions $\mathbf{t}$ before (left) and after (right) projecting out $\mathbf{t}_G$ and $\mathbf{t}_P$ from the training activations. The x-axis shows the train set and the y-axis the test set. All truth directions are trained on 80% of the data. If test and train set are the same, we evaluate on the held-out 20%, otherwise on the full test set. The displayed AUROC values are averaged over 10 training runs, each with a different train/test split.



(a) (b)

Figure 32: Gemma-2-27B: Generalization of TTPD, LR, CCS and MM. Mean and standard deviation are computed from 20 training runs, each on a different random sample of the training data.

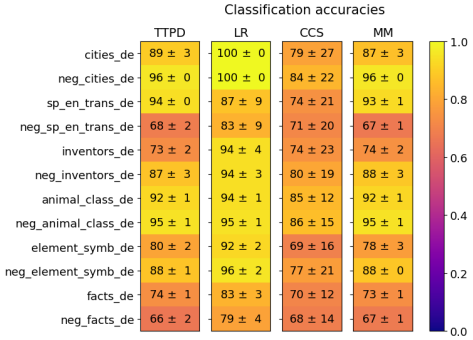


Figure 33: Gemma-2-27B: Generalization accuracies of TTPD, LR, CCS and MM on the German statements. Mean and standard deviation are computed from 20 training runs, each on a different random sample of the training data.

G.5 LLaMA3-8B-base

In this section, we present the results for the LLaMA3-8B base model. As shown in figure 34, the

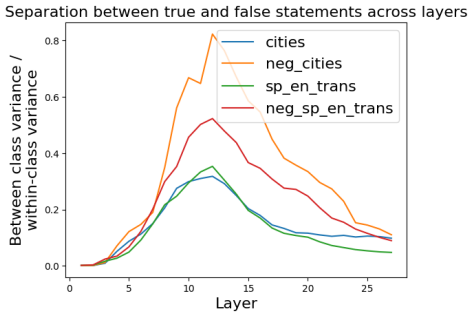
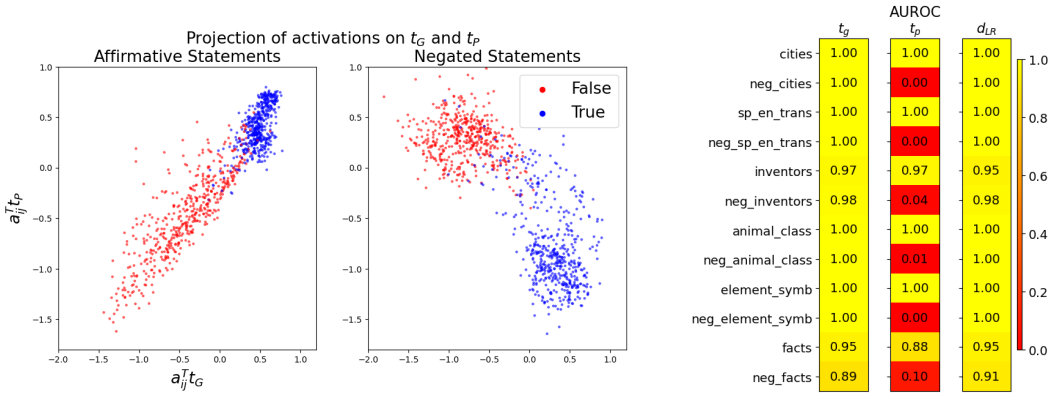


Figure 34: LLaMA3-8B-base: Ratio between the between-class variance and within-class variance of activations corresponding to true and false statements, across residual stream layers.

largest separation between true and false statements occurs in layer 12. Therefore, we use activations from layer 12 for the subsequent analysis of the LLaMA3-8B-base model.



(a) (b)

Figure 35: LLaMA3-8B-base: Left (a): Activations $\mathbf{a}_{ij}$ projected onto $\mathbf{t}_G$ and $\mathbf{t}_P$. Right (b): Separation of true and false statements along different truth directions as measured by the AUROC, averaged over 10 training runs.